TRADEFLOW · FINAL BACKTEST PHASE — FRESH-WINDOW VALIDATION

Does Phase 3 Generalize? The Go/No-Go Decision

3 windows never seen by this strategy in any prior batch, plus a direct test of removing BTC 15m from the signal pool

Generated 2026-06-22 · BTC/ETH/SOL · 15m + 1h · Config A (floor 60, fixed exit, uniform sizing)

Every Phase 3 batch (score floor, exit logic, regime sizing) was tested against the same 6 fixed windows repeatedly — real optimization risk that the strategy was tuned to those specific windows rather than general market behavior. This batch settles that question with 3 genuinely fresh windows, then asks a second, separate question on the same fresh data: does removing BTCUSDT 15m (the single worst-performing combination across Phase 3) materially improve the result, or does the lost trade volume cost more than the quality gain helps.

Window Selection — Verified Against Real Price Data, Not Memory

ID	LABEL	DATE RANGE	VERIFIED REAL BTC PRICE ACTION	OVERLAP NOTE
F1	Bull-to-correction (mixed)	2021-10-01 → 2022-03-31	$43,820 → ATH $69,000 (Nov) → $45,510 (Mar 31). Net +3.9%, but a +57%/-34% round trip.	1-month overlap with Baseline Campaign Window 3 (Oct 2021) — flagged per instructions, not substituted, since the overlap is exactly 1 month, not "more than 1 month."
F2	Choppy/volatile	2025-01-01 → 2025-06-30	$93,576 → $107,147. Net +14.5%, but with a sharp Feb-Mar crash to ~$76-78k (~-30% from January's high) before a V-shaped recovery.	Zero overlap with any prior batch window.
F3	Bear/correction	2025-10-01 → 2026-03-31	$114,049 → $68,284, a sustained -40.1% decline.	Zero overlap with any prior batch window.

The brief's own suggested substitute for the original bear candidate (2022-11→2023-04, proposed after the original 2019 candidate was ruled out for predating SOL's Aug-2020 listing) was checked and rejected — it overlaps the existing FTX-Crash-Recovery quarter (2022-11→2023-01) by a full 3 months. F3 was chosen instead, verified against real Binance price data rather than assumed, with zero overlap against anything used in this project to date.

Part 1 — Fresh-Window Validation Results

WINDOW	HAND-PICKED REGIME	CLASSIFIER'S DOMINANT LABEL	COMBINED RETURN (15M+1H)	COMBINED MAX DD	WIN RATE	PF
F1	Bull/mixed	weak_downtrend (4/6 sub-runs)	+26.4%	18.2%	56.1%	1.42
F2	Choppy	weak_downtrend (4/6 sub-runs)	+23.0%	16.9%	55.0%	1.24
F3	Bear	weak_downtrend (5/6 sub-runs)	+13.0%	16.0%	53.0%	1.23

Classifier disagreement on F1 and F2, in the same spirit as the Baseline Campaign's Window 3 finding. F1 was hand-picked as "mixed" (which the classifier's weak_downtrend lean doesn't really contradict — a topping-then-declining window). F2 is more interesting: hand-picked as "choppy" based on its violent crash-then-V-recovery character, but the classifier's per-candle sampling (taken only while flat, using a trailing 50-candle EMA window) reads most of it as weak_downtrend rather than ranging. This isn't necessarily a classifier error — a sharp crash followed by a sharp recovery genuinely produces more candles with a declining or recovering EMA structure than genuinely flat, range-bound candles. It's a reminder that "choppy" in the colloquial, narrative sense (price went nowhere net, with violent swings) doesn't always match "ranging" in the classifier's strict technical sense.

Generalization Verdict — Fresh vs. Phase 3 Baseline

METRIC	PHASE 3 (6 FIXED WINDOWS)	FRESH (3 NEW WINDOWS)	DELTA
Mean combined return	+23.6%	+20.8%	-2.85pp
Mean combined max DD	15.5%	17.0%	+1.51pp
Profitable sub-runs	56%	72%	+16pp (better)
Worst single window	-5.4% (Window 5)	+13.0% (F3)	—
Best single window	+46.0% (Window 6)	+26.4% (F1)	—

CONFIRMS. The fresh mean return (+20.8%) is within 2.85 points of the Phase 3 mean (+23.6%) — comfortably inside the ±5pp threshold. No fresh window falls below -20% or above +50%, the catastrophic-outlier bounds defined for this check. If anything, the fresh windows were slightly more consistent (no single window went negative, versus Phase 3's Window 5 at -5.4%) and the profitable-sub-run rate was actually higher (72% vs 56%). **Phase 3's improvements generalize to data the strategy has never seen.**

Part 2 — BTC 15m Removal (Variant F)

BTCUSDT 15m was Phase 3's single worst combination — negative in 4 of 6 fixed windows. This part asks the same question on the fresh data: drop BTC's 15m signals (BTC still trades on 1h, where it performs well throughout) and check whether the combined 15m+1h number improves. No new simulations were needed — each symbol/timeframe/window already ran independently in Part 1, so this is a direct re-aggregation of the same 18 runs with the 3 BTCUSDT-15m rows excluded.

CONFIGURATION	RUNS	MEAN RETURN	MEAN MAX DD	MEAN WR	PROFITABLE	TOTAL TRADES

Config A — full pool	18	+20.8%	17.0%	54.7%	13/18 (72%)	2,437
Variant F — BTC 15m removed	15	**+24.8%**	**15.7%**	**56.3%**	**12/15 (80%)**	1,853

Per-Window: A vs F

WINDOW	CONFIG A RETURN	VARIANT F RETURN	CONFIG A MAX DD	VARIANT F MAX DD
F1 (Bull/mixed)	+26.4%	+33.3%	18.2%	16.1%
F2 (Choppy)	+23.0%	+20.7%	16.9%	17.6%
F3 (Bear)	+13.0%	+20.4%	16.0%	13.5%

> **Honest caveat: the improvement is not uniform.** Window F2 is the one place removing BTC 15m makes things slightly worse (+23.0%→+20.7%, and drawdown ticks up too) — because BTC 15m specifically had its *best* result of the entire fresh batch in F2 (+34.5%, on the choppy/volatile window, the opposite of what Phase 3's pattern would predict). BTC 15m's fresh-window results were -8.1% (F1), +34.5% (F2), -24.0% (F3) — not uniformly bad the way it was across the 6 Phase 3 windows. The "BTC 15m is always a drag" pattern does not fully replicate on fresh data; it is window-dependent. Removing it still wins in aggregate (2 of 3 windows improve, and the net effect across all 3 is clearly positive), but it is not a universal fix and should be understood as a net positive on average, not a guaranteed improvement in every market condition.

Specific Checks

- **Trade volume impact:** total trades drop from 2,437 to 1,853 (-24%) — a real reduction, but 1,853 trades across 15 sub-runs (~124/run average) remains a substantial sample, not a thin one.
- **15m win rate without BTC:** 49.1% (with BTC) → 50.4% (without) — a modest +1.3pp improvement, smaller than the return improvement alone would suggest, consistent with BTC 15m dragging quality down less consistently on fresh data than it did across the Phase 3 windows.
- **Combined 15m+1h mean return:** improves from +20.8% to +24.8% (+4.0pp) — a real, if not dramatic, net positive.

Part 3 — Randomized Parameter Search

Skipped, per the brief's own scoping rule. Part 3 was conditioned on "Parts 1 and 2 both complete cleanly and the go/no-go criteria are borderline." The result below is a clean GO with comfortable margins on every criterion (return +4.79pp above threshold, drawdown 4.31pp under threshold, profitable rate 20pp above threshold) — not borderline. Running a 20-draw randomized search on top of a clean result would only add overfitting risk to the 3 fresh windows for no decision-relevant benefit, exactly the failure mode the brief's stopping rule was designed to prevent.

Go / No-Go Decision

CRITERION	THRESHOLD	VARIANT F RESULT	MET?
Fresh-window validation	CONFIRMS (±5pp, no catastrophic outlier)	-2.85pp delta, no outlier	✓ PASS
Mean return on fresh windows	≥ 20%	+24.8%	✓ PASS
Mean max drawdown on fresh windows	≤ 20%	15.7%	✓ PASS
Profitable sub-runs	≥ 60%	80%	✓ PASS

GO

All four go-criteria are met with comfortable margins, using Variant F (BTC 15m removed) as the carried-forward configuration. Fresh-window validation confirms Phase 3's improvements generalize to data this strategy has never seen — the +20.8%/+24.8% fresh numbers sit close to (not dramatically below) the Phase 3 mean of +23.6%, and the fresh windows were if anything more consistent (no negative window, higher profitable-rate). Removing BTC 15m from the signal pool is a real, net-positive lever (+4.0pp combined return, -1.3pp drawdown) even though it is not uniformly an improvement in every regime.

One honest framing point: this clears the 20%-return / 20%-drawdown / 60%-profitable GO bar defined for this specific decision, but it does not reach the original 30%-in-6-months aspiration that motivated the whole Phase 3 sequence (best result: +24.8% on fresh data). The strategy, in its current best-validated form, tops out in the mid-20s on this metric across the data tested so far — that is the honest number to carry into the infrastructure phase, not 30%.

Recommended Configuration to Carry Forward

- Symbols: ETHUSDT + SOLUSDT on 15m; BTCUSDT + ETHUSDT + SOLUSDT on 1h
- Score floor: 60
- Exit logic: fixed exit (TP1 partial close + fixed stop) — not the trailing variants, which did not beat this baseline in Phase 3B/3C
- Position sizing: uniform 1x (Config A's sizing) — not regime-conditional sizing (Condition E), which traded return for drawdown reduction in Phase 3D; worth reconsidering once the regime classifier can be monitored live, but not adopted for this go-live config
- All guardrails, HTF confirmation, session filter, regime filter: unchanged throughout

What Happens Next

This is the last backtest batch. Per the project's own scoping discipline, no further Phase 3 variants follow regardless of outcome — the strategy has been characterized thoroughly (entry quality, exit mechanics, exposure management, and now generalization) and further backtesting carries more overfitting risk than insight value. With a GO decision, the next conversation is infrastructure planning: server + database migration off Dexie/IndexedDB, a paper-live execution engine, kill switch and alerting, and a monitoring dashboard — targeting roughly 5-7 weeks to real capital, starting with paper-live testing on real prices before any money is at risk.

Appendix — All 18 Fresh-Window Sub-Run Results

WINDOW	SYMBOL	TF	TRADES	WR	PF	RETURN	MAX DD	NOTE
F1	SOLUSDT	1h	60	62.7%	1.63	+37.6%	9.3%	
F1	SOLUSDT	15m	211	47.9%	1.08	+11.9%	33.5%	
F1	ETHUSDT	1h	67	72.7%	2.42	+61.3%	6.1%	
F1	ETHUSDT	15m	210	53.3%	1.23	+42.6%	18.8%	
F1	BTCUSDT	1h	83	52.4%	1.18	+12.9%	12.6%	
F1	BTCUSDT	15m	203	47.3%	0.95	-8.1%	29.1%	Excluded in F
F2	SOLUSDT	1h	72	63.9%	1.89	+61.7%	17.0%	
F2	SOLUSDT	15m	226	54.4%	1.26	+51.9%	12.5%	
F2	ETHUSDT	1h	84	52.4%	0.93	-5.4%	18.2%	
F2	ETHUSDT	15m	190	48.9%	0.82	-20.9%	26.8%	
F2	BTCUSDT	1h	68	57.4%	1.30	+16.2%	13.2%	
F2	BTCUSDT	15m	190	53.2%	1.26	+34.5%	13.5%	Excluded in F
F3	SOLUSDT	1h	74	58.1%	1.39	+27.2%	13.3%	
F3	SOLUSDT	15m	186	44.6%	0.92	-9.0%	19.3%	
F3	ETHUSDT	1h	65	60.9%	1.44	+25.1%	6.4%	
F3	ETHUSDT	15m	202	53.0%	1.16	+29.3%	17.4%	
F3	BTCUSDT	1h	55	61.8%	1.71	+29.6%	10.9%	
F3	BTCUSDT	15m	191	39.3%	0.77	-24.0%	28.8%	Excluded in F

Rows shown at reduced opacity are BTCUSDT 15m — excluded from Variant F's aggregation, included in Config A's.

Report generated from 18 newly-run live simulations against real historical Binance market data spanning genuinely fresh date ranges (Oct 2021-Mar 2022, Jan-Jun 2025, Oct 2025-Mar 2026). Zero console errors. Raw data: `training-exports/fresh-validation-configA-results.json`.